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Primary Knee

Natural Language Processing of Patient-Experience Comments After Primary Total Knee Arthroplasty



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ABSTRACT

Background: There is interest in improving patient experience after total knee arthroplasty (TKA) due to recent shifts toward value-based medicine. Patient narratives are a valuable but unexplored source of information.

Methods: Records of 319 patients who had undergone primary TKA between August 2016 and August 2019 were linked with vendor-supplied patient satisfaction data, which included patient comments and the Press Ganey satisfaction survey. Using machine-learning-based natural language processing, 1048 patient comments were analyzed for sentiment and classified into themes. Postoperative outcomes, patient-reported outcome measures, and traditional measures of satisfaction were compared between patients who provided a negative comment vs those who did not (positive, neutral, mixed grouped together). Multivariable regression was used to determine perioperative variables associated with providing a negative comment.

Results: Of the 1048 patient comments, 25% were negative, 58% were positive, 8% were mixed, and 9% were neutral. Top 2 themes of negative comments were room condition (25%) and inefficient communication (23%). There were no differences in most of the studied outcomes (eg, peak pain intensity, length of stay, or Knee Injury and Osteoarthritis Outcome Score Junior and pain scores at 6-week follow-up) between the 2 cohorts ($P > .05$). However, patients who made negative comments were less likely to highly recommend their hospital care to peers ($P < .001$). Finally, patients who had higher American Society of Anesthesiologists Score and those who received a scopolamine patch were more likely to provide negative comments ($P < .05$).

Conclusion: Although the current study showed that patient satisfaction might not be a proxy for traditional objective perioperative outcomes, efforts to improve the nontechnical aspects of medicine are still crucial in providing patient-centered care.

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With a recent shift in orthopedic surgery from fee-for-service to value-based care, enhancing patient experience has become a strategic approach for achieving high-value musculoskeletal care [1]. The importance of measuring and reporting on patient experience has been underscored by the inclusion of the Hospital Consumer Assessment of Healthcare Provider and Systems (HCAHPS) survey scores in the Centers for Medicare and Medicaid Services Value-Based Purchasing Program, as payers are increasingly linking patient satisfaction measures to reimbursement [2,3]. However, a growing number of studies have questioned the validity of utilizing patient satisfaction scores as a measure of provider quality, noting inconsistent associations with clinical and functional outcomes [4–7].

The HCAHPS survey has been widely adopted as a standardized measurement of patient satisfaction with the delivery of care [6,8]. The survey uses close-ended questions to assess patients' perception of their care across several domains. However, due to the close-ended nature, the questions are limited in their ability to capture the full range of patient experiences. Patients are able to provide additional information in the form of free-text comments, but are not factored into overall ratings, and their content remains largely unexplored [9].

With the advent of machine-learning-based natural language processing, analysis of large amounts of unstructured free-text data has now become possible. Although its use has been limited in healthcare [10,11], other industries have been quick to adopt this innovative methodology. For example, in the finance sector, natural language processing has been used to extract free-text data from online newsfeeds to predict volatility in stock prices [12]. Furthermore, other consumer services, such as Netflix, Amazon, and Spotify, have embraced this technology to predict user preferences [13].

Therefore, the purposes of this study were to employ machine-learning-based natural language processing to (1) determine common content of negative patient-experience comments after primary total knee arthroplasty (TKA), (2) identify any differences in postoperative outcomes, patient-reported outcome measures (PROMs), and satisfaction between patients who provided a negative comment vs those who did not, and (3) test for an association between perioperative variables and providing a negative comment.

Patients and Methods

Patient Population

This retrospective study was performed at an urban orthopedic specialty hospital in the United States. After Institutional Review Board approval, an institutional registry was used to link the records of patients who had undergone elective primary TKA between August 2016 and August 2019 to vendor-supplied patient satisfaction data, which included free-text patient-experience comments and the Press Ganey satisfaction survey. Patients who were included in the study had their procedure performed by either of 2 fellowship-trained arthroplasty surgeons. Patients who did not complete a postoperative Press Ganey satisfaction survey and provided comments, approximately 50% of all primary TKA cases performed by these 2 surgeons during this time period, were excluded. In order to ensure a homogenous population, patients who underwent revision surgeries were excluded. This yielded a total cohort of 319 patients.

Patient-Experience Comments

With use of machine-learning-based natural language processing developed by the survey vendor (Press Ganey Associates), all patient comments were analyzed for sentiment and classified as positive, negative, mixed, or neutral [14]. Machine learning is a novel field in computer science in which a model learns from a large set of examples (eg, here is a negative comment and here is a positive comment) rather than being programmed with explicit rules [15]. Natural language processing is a form of machine learning used to analyze and understand the meaning of vast amounts of free text [16]. Natural language processing can be used to analyze emotion or sentiment within a free-text phrase [9,17].

All comments were also manually reviewed to confirm the accuracy of the automated sentiment analysis. In addition, to provide more insight, the negative comments were manually classified into

themes relating to (1) room condition, (2) time management, (3) communication, (4) compassion, (5) difficult IV (intravenous) insertion, (6) food, (7) medication side effects, (8) discharge instructions, and (9) pain management. These classifications were based on previous literature [9].

Perioperative Variables

Preoperative Characteristics

Several patient preoperative characteristics were collected for the current study. These included age, gender, body mass index, insurance type, American Society of Anesthesiologists (ASA) score, and number of patient-reported allergies. Higher numbers of patient-reported allergies have been associated with lower patient satisfaction [18].

Operative Characteristics

Operative variables collected for the current study included operative time (in minutes) and anesthesia type. Anesthesia type was categorized as general or regional.

Postoperative Outcomes

Several postoperative outcomes were assessed. These included length of stay (measured in days), non-homebound discharge, received blood transfusion, received scopolamine patch, and received straight catheterization. Other variables such as dosage of ondansetron received (IV or oral), inpatient peak pain intensity, and inpatient opioid intake (measured in oral morphine equivalents) were also included.

Patient-Reported Outcome Measures

Preoperative Knee Injury and Osteoarthritis Outcome Scores Junior (KOOS Jr) and preoperative numeric pain rating scale scores were collected. In addition, postoperative KOOS Jr scores and numeric pain rating scale at 6-week follow-up were also included. However, these data were only available for a subset of patients (66% for KOOS Jr and 60% for numeric pain rating scale). The KOOS Jr provides a single score representing "knee health" as it combines pain, symptoms, and functional limitations in a single score. This short-form PROM is patient-relevant, efficient, and validated [19]. The KOOS Jr scoring is on a 100-point scale with 0 representing complete knee disability and 100 representing perfect knee health, just as with the original KOOS [19]. The numeric pain rating scale is a patient-reported measure where 0 represents no pain and 10 represents the maximum pain level.

Patient Satisfaction

Overall satisfaction with the process of care was evaluated with the Press Ganey patient satisfaction survey. This survey is administered to inpatients between 48 hours and 6 weeks after hospital discharge to capture multiple aspects of the hospital experience. Specifically, we utilized the "overall rating of hospital care" domain to evaluate association of negative patient-experience comments with more traditional methods of assessing patient satisfaction. We considered the patient to be "satisfied" with their care if they chose the most positive or "top-box" item for the section "Would you recommend this hospital to your friends and family?"

Statistical Analysis

Patient preoperative characteristics, operative characteristics, postoperative outcomes, PROMs, and Press Ganey patient satisfaction survey response were compared using the Pearson chi-

Table 1
Preoperative Characteristics of the Study Population.

Total	Negative Comments		P-Value ^a
	Yes	No	
	Number = 154	Number = 165	
Age (y)	Average: 63.1 ± 6.7	Average: 64.8 ± 7.7	.315
<55	18 (11.69%)	18 (10.91%)	
55–64	72 (46.75%)	71 (43.03%)	
65–74	54 (35.06%)	55 (33.33%)	
>74	10 (6.49%)	21 (12.73%)	
Gender			.165
Male	43 (27.92%)	58 (35.15%)	
Female	111 (72.08%)	107 (64.85%)	
Body mass index (kg/m ²)	Average: 31.7 ± 7.8	Average: 30.1 ± 6.1	.221
18–25	32 (20.78%)	33 (20.00%)	
25–30	38 (24.68%)	52 (31.52%)	
30–35	39 (25.32%)	47 (28.48%)	
>35	45 (29.22%)	33 (20.00%)	
Insurance type			.614
Medicare	23 (14.94%)	30 (18.18%)	
Private	129 (83.77%)	134 (81.21%)	
Other (Worker's Compensation, Medicaid, etc.)	2 (1.30%)	1 (0.61%)	
American Society of Anesthesiologists Classification	Average: 2.2 ± 0.5	Average: 2.1 ± 0.4	.067
1–2	115 (74.68%)	137 (83.03%)	
3	39 (25.32%)	28 (16.97%)	
Number of allergies			.310
0–1	102 (66.23%)	106 (64.24%)	
2	22 (14.29%)	17 (10.30%)	
>2	30 (19.48%)	42 (25.45%)	

^a Chi-squared tests or Student's *t*-tests were used to compare these variables (significant at *P* < .05).

squared tests for categorical variables and the independent samples *t*-tests for continuous variables between patients who provided a negative comment vs those who did not. In addition, change in mean KOOS Jr scores and change in mean numeric pain rating scale scores between preop and 6-week follow-up were also compared using independent samples *t*-tests.

The occurrence of a negative comment was tested for association with each of the aforementioned preoperative, operative, and postoperative variables in Tables 1–3 using Poisson regression with robust error variance. The final multivariate model was selected using a backward stepwise approach, where all variables in Tables 1–3 were initially included in the model and variables with the highest *P* values were eliminated one by one until only variables with *P* < .05 remained in the model. Variables remaining in the model represented independent associations with providing a negative comment.

Statistical tests were 2-sided, and the threshold for a type I error (alpha) was set at 0.05. All statistical analyses were performed in Stata version 13.1 (StataCorp, LP, College Station, TX).

Results

Patient Population

A total of 319 patients met the previously described inclusion criteria. Mean age for the entire study population was 64.0 ± 7.3 years, 68.3% of patients were female, mean body mass index was 30.9 ± 7.0 kg/m², 82.4% of patients had private insurance, mean ASA score was 2.2, and the average number of allergies was 1.6 (Table 1). In addition, the average operative time was 91.6 minutes and 98.1% of cases used regional anesthesia (Table 2). Of the 319 patients, 154 patients provided at least one negative comment while 165 patients did not (provided positive, neutral, or mixed grouped together). There were no statistical differences in number of patient-reported allergies between patients who provided a negative comment vs those who did not (*P* = .310).

Patient-Experience Comments

Response rate of the Press Ganey patient satisfaction survey for all patients who underwent primary TKA was approximately 50%. A total of 319 patients who underwent primary TKA and completed the Press Ganey patient satisfaction survey were identified. Overall, there were 1084 comments, of which 25% were negative, 58% were positive, 8% were mixed, and 9% were neutral (Fig. 1). All comments were also manually reviewed to confirm the accuracy of the automated sentiment analysis. The agreement rate was excellent at 96.3%. The most common theme of the negative comments was related to room condition (25%), followed by inefficient communication (23%), lack of compassion (15%), time management (10%), discharge instructions (10%), food (7%), pain management (5%), difficult IV insertion (4%), and medication side effects (2%) (Fig. 2). The negative comments related to room condition mostly focused around these major topics of dissatisfaction: loud roommates, excessive noise, uncomfortable temperature, and insufficient space for visitors. The negative comments regarding inefficient communication mostly centered on insufficient explanation of postsurgical treatment plans and poor explanation of different forms of documentation.

Table 2
Operative Characteristics of the Study Population.

Total	Negative Comments		P-Value ^a
	Yes	No	
	Number = 154	Number = 165	
Operative time (min)			.243
<75	44 (28.57%)	50 (30.30%)	
75–99	43 (27.92%)	33 (20.00%)	
>99	67 (43.51%)	82 (49.70%)	
Anesthesia type			.460
General	2 (1.30%)	4 (2.42%)	
Regional	152 (98.70%)	161 (97.58%)	

^a Chi-squared tests were used to compare these variables (significant at *P* < .05).

Table 3
Postoperative Outcomes of the Study Population.

Total	Negative Comments		P-Value ^a
	Yes	No	
	Number = 154	Number = 165	
Length of stay (d)	Average: 2.6 ± 0.8	Average: 2.6 ± 1.0	.799
1–2	79 (51.30%)	87 (52.73%)	
3	75 (48.70%)	78 (47.27%)	
Non-homebound discharge			.192
No	148 (96.10%)	153 (92.73%)	
Yes	6 (3.90%)	12 (7.27%)	
Received blood transfusion			.945
No	152 (98.70%)	163 (98.79%)	
Yes	2 (1.30%)	2 (1.21%)	
Scopolamine patch			.015
No	132 (85.71%)	155 (93.94%)	
Yes	22 (14.29%)	10 (6.06%)	
Straight catheterization			.880
No	145 (94.16%)	156 (94.55%)	
Yes	9 (5.84%)	9 (5.45%)	
Peak pain intensity	Average: 7.5 ± 1.7	Average: 7.2 ± 1.8	.059
0–4	8 (5.19%)	15 (9.09%)	
5–8	98 (63.64%)	116 (70.30%)	
9–10	48 (31.17%)	34 (20.61%)	
Inpatient opioid intake (oral morphine equivalents)	Average: 126.6 ± 61.7	Average: 117.7 ± 62.5	.204
Ondansetron (PO, mg)	Average: 0.9 ± 2.6	Average: 0.5 ± 1.6	.123
Ondansetron (IV, mg)	Average: 10.8 ± 4.7	Average: 11.1 ± 4.2	.515

Bold value indicates statistical significance.

IV, intravenous.

^a Chi-squared tests or Student's *t*-tests were used to compare these variables (significant at *P* < .05).

Postoperative Outcomes, Patient-Reported Outcome Measures, and Patient Satisfaction

There were no identified statistical differences in most of the studied postoperative outcomes between patients who provided a negative comment vs those who did not (positive, neutral, mixed grouped together) in the current study. These outcomes included length of stay, non-homebound discharge, rate of blood transfusion, and rate of straight catheterization (*P* > .05) (Table 3). In addition, there were no statistical differences in patient-reported peak pain intensity during their hospitalization and in inpatient opioid intake (measured in oral morphine equivalents) between the 2 groups (*P* > .05) (Table 3). Furthermore, there were also no differences in ondansetron dosages (IV or PO) received (*P* > .05) (Table 3). However, patients who provided a negative comment

tended to have received a scopolamine patch (14.3% vs 6.1%, *P* = .015) (Table 3).

In terms of PROMs, there were no statistical differences in preop KOOS Jr scores and preop numeric pain rating scale scores between patients who provided a negative comment vs those who did not (*P* > .05) (Table 4). In addition, there were no differences in 6-week KOOS Jr and numeric pain rating scores (*P* > .05) (Table 4). Finally, there were also no statistical differences in change in mean KOOS Jr scores and change in mean numeric pain rating scale scores between preop and 6-week follow-up (*P* > .05) (Fig. 3).

However, patients who made negative comments were less likely to be “satisfied” with their care since they were less likely to choose the most positive or “top-box” item for the section “Would you recommend this hospital to your friends and family?” (78.6% vs 93.9%, *P* < .001) (Table 4).

Associations Between Perioperative Variables and Negative Comments

In the multivariate analysis controlling for each perioperative variable in Tables 1–3, independent risk factors for leaving a negative comment included higher ASA scores (ASA = 3 relative to ASA = 1–2, relative risk = 1.32, 95% confidence interval = 1.04–1.69, *P* = .025) and those who received a scopolamine patch (relative risk = 1.37, 95% confidence interval = 1.17–1.60, *P* < .001) (Table 5). There were no statistical associations between number of patient-reported allergies and leaving a negative comment.

Discussion

Currently, efforts to measure patient satisfaction rely primarily on close-ended survey items that are limited in their ability to capture the full range of patient experiences. Free-text patient comments remain a largely unexplored, but potentially valuable source of information that can help guide best practices. This study investigated the content of negative patient-experience comments

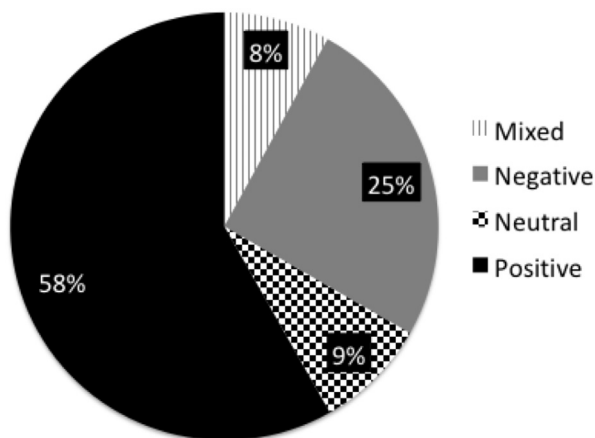


Fig. 1. Distribution of sentiment of patient-experience comments after primary total knee arthroplasty. Each comment is categorized as positive, negative, mixed, or neutral.

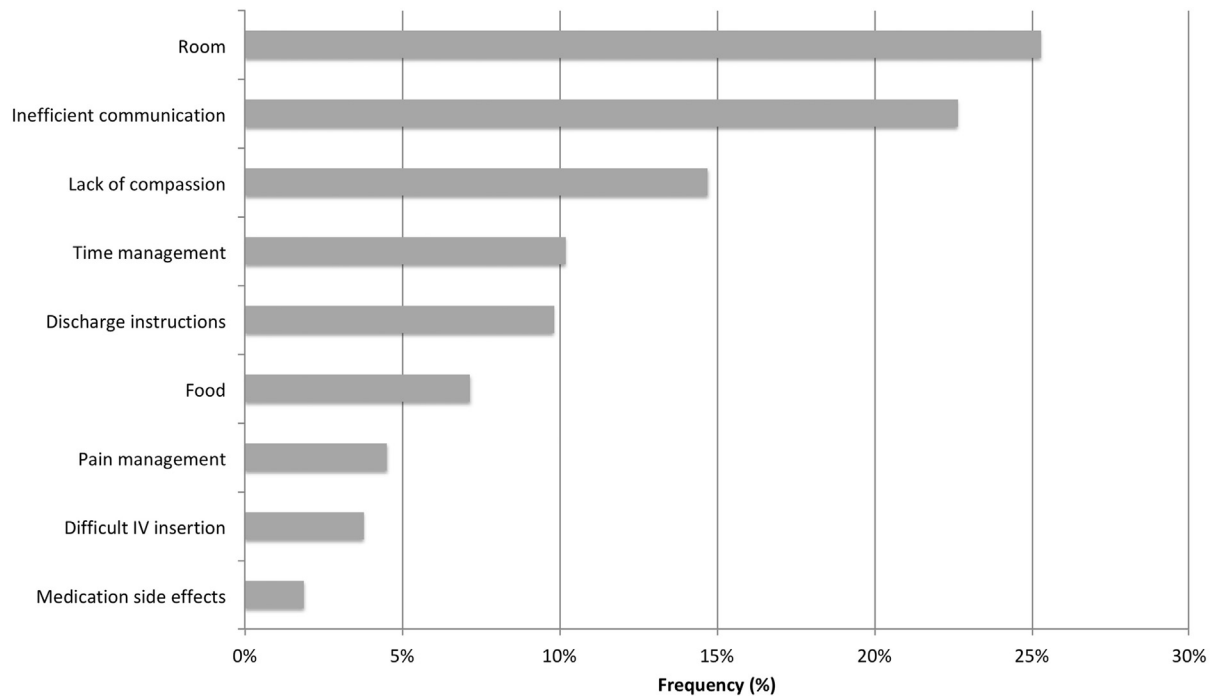


Fig. 2. Distribution of topics of negative comments. Each negative comment is categorized as being related to room condition, inefficient communication, lack of compassion, time management, discharge instructions, food, pain management, difficult intravenous (IV) insertion, or medication side effects.

after primary TKA, along with postoperative outcomes, and associated perioperative variables. The 2 main themes for negative comments were room conditions and inefficient communication. There were no differences in outcomes such as inpatient peak pain intensity, inpatient opioid intake, length of stay, discharge disposition, or in 6-week postoperative KOOS Jr and pain scores between those who provided negative comments vs those who did not. However, patients who made negative comments were less likely to recommend their hospital care to their peers. Finally, patients who received a scopolamine patch and those with higher ASA scores were more likely to provide negative comments.

The most common theme for patient dissatisfaction was room conditions (25%). This finding is consistent with a similar study of a cohort of patients who underwent total shoulder arthroplasty, where 27% of the negative comments were also attributed to room conditions [9]. Although there is growing emphasis on the doctor-patient relationship for improving patient experience, focusing on that alone fails to recognize the physical hospital environment that patients seem to consider important [20–22]. In a previous study,

investigators implemented a “quiet time” program during designated hours by dimming light, asking staff to lower their voices, and replacing noisy cart wheels, and subsequently observed an improvement in patient satisfaction scores [23]. Thus, limiting noise pollution, improving spatial layouts, providing additional seating for family, and allowing for private rooms can be effective strategies to enhance room conditions and drive an overall improvement in patient satisfaction [24].

Among negative comments, the second most common theme was inefficient communication between patients and their care teams. Previous studies of HCAHP survey domains have found that overall hospital ratings after total joint arthroplasty (TJA) were significantly influenced by communication and responsiveness with nurses, doctors, and other hospital staff [25,26]. Inefficient communication has also been associated with increased pain intensity immediately after TKA [27], suggesting that effective communication can play a role in managing pain symptoms. Furthermore, effective communication between the patient and the care team can also help engage patients and promote discussions

Table 4
Patient-Reported Outcome Measures and Patient Satisfaction.

Total	Negative Comments		P-Value ^a
	Yes	No	
	Number = 154	Number = 165	
Preop KOOS Jr	Average: 50.1 ± 13.28	Average: 48.6 ± 14.2	.426
Preop Numeric Pain Rating Scale	Average: 5.9 ± 2.5	Average: 5.8 ± 2.5	.745
KOOS Jr 6 weeks postop	Average: 59.3 ± 11.5	Average: 62.1 ± 11.9	.084
Numeric Pain Rating Scale 6 weeks postop	Average: 4.0 ± 2.7	Average: 3.6 ± 2.3	.237
Extremely likely to recommend to peers			<.001
No	33 (21.43%)	10 (6.06%)	
Yes	121 (78.57%)	155 (93.94%)	

Bold value indicates statistical significance.

KOOS Jr, Knee Injury and Osteoarthritis Outcome Score Junior.

^a Chi-squared tests or Student's *t*-tests were used to compare these variables (significant at *P* < .05).

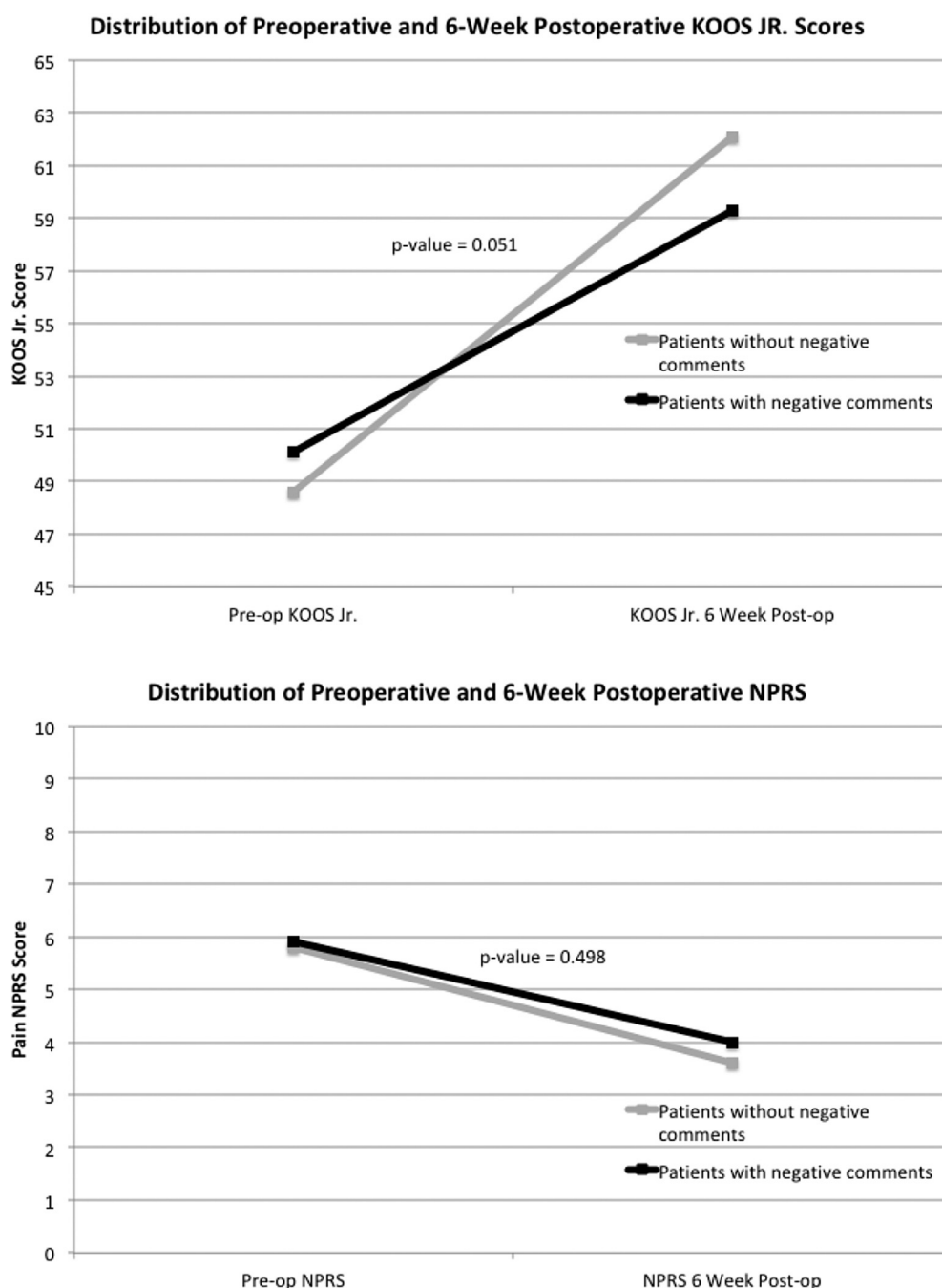


Fig. 3. Distribution of mean preoperative and mean 6-week postoperative KOOS Jr scores. The *P*-value represents comparison of change in mean KOOS Jr scores at preop and 6-week follow-up between patients who had a negative comment vs those who did not. Distribution of mean preoperative and mean 6-week postoperative NPRS scores. The *P*-value represents comparison of change in mean NPRS scores at preop and 6-week follow-up between patients who had a negative comment vs those who did not. KOOS Jr, Knee Injury and Osteoarthritis Outcome Score Junior; NPRS, Numeric Pain Rating Scale.

related to expectations during postoperative recovery, which has been correlated to improved satisfaction across various orthopedic procedures including TKA [28–34]. The current findings support that while surgical experience and technical skills are important, patients also seem to value interpersonal skills such as communication, empathy, respect, and professionalism from their care team [35,36].

This study found no differences in traditional measures of clinical outcomes such as peak pain intensity, opioid intake, length of stay, discharge disposition, or improvements in PROMs between those who provided negative comments vs those who did not.

However, patients who made negative comments were less likely to recommend their hospital care to peers, which highlights an inconsistency between the factors that influence how patients rate their hospital experience and their eventual functional outcomes [6,37]. This is echoed by several studies that question the use of patient experience measures to assess provider quality and determine reimbursement [4,5,25,38–42]. By analyzing free-text negative patient comments, this study identified specific themes like room conditions (eg, number of additional chairs in the room, room temperature) that seem unrelated to functional recovery, but influence negative ratings. Surveys that wish to appropriately assess

Table 5
Independent Risk Factors for Negative Comments.

Characteristic	RR	95% CI	P-Value ^a
American Society of Anesthesiologists Classification			.025
1–2	Reference	—	
3	1.32	1.04–1.69	
Scopolamine patch			<.001
No	Reference	—	
Yes	1.37	1.17–1.60	

Bold values indicate statistical significance.

RR, relative risk; CI, confidence interval.

^a Significant at $P < .05$.

provider quality with patient satisfaction will need to consider which domains are being evaluated and whether they appropriately reflect the care provided.

In terms of predictors of leaving a negative comment, patients who made negative comments were more likely to have received a scopolamine patch. A prospective study of 10,811 patients undergoing surgery found a strong correlation between patient dissatisfaction and postoperative nausea and vomiting [43]. In the current study, it is possible that patients who received scopolamine patches were experiencing increased nausea and vomiting symptoms that negatively impacted their postoperative recovery. Postoperative nausea and vomiting can interfere with rehabilitation therapy after surgery and thereby affect functional recovery, increase length of stay, increase cost of treatment, and lead to other systemic complications [43–45]. Appropriate use of antiemetics as part of a multimodal pain management in patients undergoing TKA can play a role in enhancing patient experience.

Finally, patients who had a higher prevalence of medical comorbidities (ASA score) were more likely to leave a negative comment. A study of 37,223 hospitalized patients found that a higher comorbidity burden was inversely associated with HCAHPS patient satisfaction scores [46]. Among TJA patients, a higher number of comorbidities has been previously associated with higher rates of dissatisfaction [9,47]. It is possible that sicker patients have more complex medical needs, which may necessitate placement in busier monitored settings or require attention that detracts from their primary goal of surgical recovery, leading to poor perceived patient experiences. Furthermore, sicker patients may be unable to achieve the same benefits from surgery due their comorbid conditions, and therefore remain less satisfied than their healthier peers. However, as healthcare shifts from a fee-for-service to value-based model, where providers are incentivized to improve patient satisfaction scores, there is potential to introduce behaviors that specifically focus on improving survey scores without the best interest of the patient [48,49]. Without appropriate risk adjustment for complex patients, “cherry picking” (strategic selection of low risk patients) and “lemon dropping” (denying care to high risk patients) are likely to occur in order to improve satisfaction scores and overall hospital ratings [49,50]. This will potentially limit access to TJA for at-risk populations and will concentrate their care to select safety net hospitals, further exacerbating disparities in care delivery [51,52]. There is optimism in recent literature that shows no significant change in the characteristics of patients undergoing TJA after the introduction of bundled payment policies, but these findings tell us little about patients who may have been appropriate candidates for joint replacement but did not undergo the procedure due to higher health risks [53]. Recognizing patients with higher prevalence of comorbidities and optimizing modifiable risk factors will be important steps toward patient safety and potentially enhancing patient experience. At the same time, appropriate risk adjustment for complex patients when reviewing satisfaction scores will be

important for protecting equitable access to TJA. There were no statistical associations between other characteristics such as number of patient-reported allergies and providing a negative comment.

This study was subject to certain limitations. First, the data were limited to a single high-volume orthopedic specialty hospital and may lack generalizability. Second, selection and nonresponse biases must be considered, because patients with exceptionally poor or overwhelmingly positive experiences are more likely to respond to surveys [54,55]. Furthermore, limitations include the use of 6-week postoperative PROMs rather than longer term outcomes, but a previous study supports that most improvement after TKA occurred within the first 3 months with no significant change thereafter [56]. In addition, only approximately 50% of all patients who underwent primary TKA by these 2 surgeons during this time period completed the postoperative Press Ganey satisfaction survey. Based on how survey data were collected, it was not possible to identify patients who did not complete the Press Ganey satisfaction survey to analyze any demographic differences. Furthermore, other important factors that may influence patient satisfaction, such as their level of health literacy, expectations, engagement, or knee-specific factors such as type of deformity (varus/valgus), severity of deformity, range of motion, and prior surgery, were unavailable given the retrospective nature of this study. Finally, timing of completion for each Press Ganey satisfaction survey was not recorded and the effects of timing on providing negative comments could not be analyzed.

Conclusion

In conclusion, exploring the content of negative patient-experience comments can allow for meaningful insight into aspects of the patient experience that drive responses to questions regarding patient satisfaction. The current study showed that patient satisfaction might not be a proxy for traditional objective postoperative outcomes and PROMs, thus calling into question the use of these subjective metrics to determine “quality” and “value” of care. However, efforts to enhance hospital conditions, refine communications, and improve the nontechnical aspects of medicine can still play a crucial role in delivering high-quality, patient-centered care after primary TKA.

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